Title of Experiment: Implementation of OLAP operations: Slice, Dice, Rollup, Drilldown and Pivot for the above problem statement in (experiment 1).
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Topic : Experiment 2 - DWM
Github Link : https://github.com/yashgrathod/DWM-Experiments

Aim:

Implementation of OLAP operations: Slice, Dice, Rollup, Drilldown and Pivot for the above problem statement in (experiment 1).

Theory:

OLAP (Online Analytical Processing) is used in data warehouses to analyze large amounts of data from multiple dimensions like time, location, and weather parameters (temperature, pressure, rainfall).

- Slice: Selects data by fixing one dimension (e.g., weather data for a specific year).
- Dice: Filters data using multiple conditions (e.g., specific time, location, and parameter).
- Roll-up: Aggregates data to a higher level (e.g., hourly data to monthly averages).
- Drill-down: Shows detailed data by going to a lower level (e.g., yearly to daily data).
- Pivot: Rotates data to view it from different perspectives (e.g., swapping rows and columns).

These operations help in easy analysis and better decision-making using large weather datasets.

Code:

```python
import pandas as pd
import numpy as np

np.random.seed(0)

data = {
    "probe_id": np.random.randint(1, 6, 100),
    "location": np.random.choice(["Land", "Ocean"], 100),
    "region": np.random.choice(["North", "South", "East", "West"], 100),
    "year": np.random.choice([2022, 2023, 2024], 100),
    "month": np.random.randint(1, 13, 100),
    "day": np.random.randint(1, 29, 100),
    "hour": np.random.randint(0, 24, 100),
    "temperature": np.random.uniform(15, 40, 100),
    "pressure": np.random.uniform(980, 1050, 100),
    "precipitation": np.random.uniform(0, 20, 100)
}

df = pd.DataFrame(data)
```

```python
print("\nOriginal Data:")
print(df.head())

print("\nSLICE: Data for year = 2023")
slice_df = df[df["year"] == 2023]
print(slice_df.head())

print("\nDICE: Year=2023, Region=North")
dice_df = df[(df["year"] == 2023) & (df["region"] == "North")]
print(dice_df.head())

print("\nROLL-UP: Avg temperature by Region")
rollup_df = df.groupby("region")["temperature"].mean()
print(rollup_df)

print("\nDRILL-DOWN: Avg temperature by Region and Month")
drilldown_df = df.groupby(["region", "month"])["temperature"].mean()
print(drilldown_df)

print("\nPIVOT: Avg temperature (Region vs Year)")
pivot_df = pd.pivot_table(df,
                          values="temperature",
                          index="region",
                          columns="year",
                          aggfunc="mean")
print(pivot_df)
```

Output:

```
SLICE: Data for year = 2023
    probe_id location region  year  month  day  hour  temperature     pressure  precipitation
3          4    Ocean   West  2023      4   20     4    21.035466   981.701924       1.252720
4          4     Land  North  2023     10   13    17    31.562614  1003.982769       4.838034
5          2    Ocean  North  2023      3   19    18    21.151580  1023.556174       8.645630
12         3     Land   West  2023      8   13    23    25.371422  1046.420271      14.907961
16         2    Ocean  South  2023      7   18    15    16.150183  1021.211773       7.996034

DICE: Year=2023, Region=North
    probe_id location region  year  month  day  hour  temperature     pressure  precipitation
4          4     Land  North  2023     10   13    17    31.562614  1003.982769       4.838034
5          2    Ocean  North  2023      3   19    18    21.151580  1023.556174       8.645630
20         4    Ocean  North  2023     12   27    14    39.637286  1036.957505      13.135239
21         1     Land  North  2023      1    5    16    39.224293  1015.686504      17.241252
23         1     Land  North  2023      4    6     8    22.413907  1035.358811       9.955538

ROLL-UP: Avg temperature by Region
region
East     25.530486
North    29.743150
South    26.929025
West     28.714489
Name: temperature, dtype: float64
```

```
DRILL-DOWN: Avg temperature by Region and Month
region  month
East    1       35.374162
        2       24.397619
        3       31.606389
        4       28.496922
        5       15.197103
        7       22.617123
        9       29.426964
        10      24.540468
        11      27.094575
        12      15.755881
North   1       33.305687
        2       30.559611
        3       29.828411
        4       29.196817
        5       23.542453
        6       28.867195
        7       32.244207
        8       27.646660
        9       24.799138
        10      30.832780
        11      28.591951
        12      39.637286
South   1       24.357405
        2       28.285064
        3       27.932713
        4       28.494088
        5       25.205263
        6       17.615697
        7       20.233580
        8       24.476422
        10      32.403248
        11      29.219692
        12      31.282091
West    1       29.670957
        2       24.325872
        4       21.035466
        5       19.268554
        6       30.839071
        7       20.945181
        8       30.234906
        9       29.660176
        10      29.109437
        11      28.723967
        12      33.622614
Name: temperature, dtype: float64
```

```
PIVOT: Avg temperature (Region vs Year)
year         2022       2023       2024
region
East    24.771200  26.234039  25.848600
North   28.419886  31.236580  26.907422
South   25.215042  24.740538  32.300624
West    28.822785  27.686609  29.480808
```

Conclusion:

The implementation of OLAP operations on the weather data warehouse successfully demonstrated how large volumes of multidimensional data can be efficiently analyzed. By applying Slice and Dice operations, specific subsets of weather data were easily filtered based on time and location. Roll-up enabled aggregation of detailed hourly data into higher-level summaries such as monthly or regional averages, while Drill-down allowed deeper analysis by exploring more granular data levels. Pivot operation further enhanced data visualization by restructuring the dataset for better comparison and interpretation.

Overall, this experiment highlights the importance of OLAP techniques in transforming raw weather data into meaningful insights, supporting faster analysis and effective decision-making in a regional weather bureau.

Review Questions:

Q1.What is the difference between the Slice and Dice operations in OLAP?

Ans.

Feature	Slice Operation	Dice Operation
Definition	Selects data by fixing one dimension	Selects data using multiple conditions
Dimensions Used	Only one dimension	Multiple dimensions
Result	Reduces cube by one dimension	Creates a smaller sub-cube
Filtering	Single filter condition	Multiple filter conditions
Example	Data for year 2025 only	Data for Jan–March 2025, coastal areas, temperature
Complexity	Simple	More complex than slice

Q2.How does the Roll-up operation help in summarizing large volumes of data?

Ans. The Roll-up operation is used to summarize large volumes of data by aggregating detailed data into higher-level information based on a hierarchy (such as time or location).

- It combines smaller data points (e.g., hourly readings) into larger summaries (daily, monthly, yearly).
- This reduces data size, making analysis faster and more efficient.
- It helps in identifying overall trends, patterns, and comparisons across different periods or regions.
- It supports better decision-making by presenting a simplified view instead of raw detailed data.
- Roll-up can be applied on different dimensions like:
    - Time: Hour → Day → Month → Year
    - Location: City → State → Region → Country

Example:
In a weather data warehouse, thousands of hourly temperature readings from probes can be rolled up into monthly or yearly averages, making it easier to study climate trends over time.

Q3.Give an example of a scenario where Pivoting the data provides a clearer insight than traditional tabular view?

Ans. In a weather data warehouse, suppose you have data like:

Location	Month	Temperature
Mumbai	Jan	25°C
Delhi	Jan	15°C
Mumbai	Feb	27°C
Delhi	Feb	18°C

Traditional Tabular View Problem:

- Data is listed row by row
- Hard to quickly compare temperatures across locations for each month After Pivoting:

Month	Mumbai	Delhi
Jan	25°C	15°C
Feb	27°C	18°C

Pivoting rearranges data (rows ↔ columns) to make comparisons and patterns more clear than a traditional tabular format.